Update your deployment

Your MongoDB cluster, Render service and Android Studio project stay exactly as they are. This update replaces the placeholder screens with the working ones, and the skeleton backend with the implemented one.

Today	After this update
Backend: `/api/health` only, every other route returns HTTP 501	Ten working endpoints: query, knowledge, schemes, scholarships, reminders, scam check, SMS, voice, sync
App: seven screens read "Coming soon"	Assistant, Health, Schemes, Scholarships, Reminders, Scam Checker all live
App: no SMS or voice entry points	Home gains SMS Centre and Virtual Call tiles (8 tiles total)
App: `SyncWorker` logs a count and stops	SyncWorker drains the Room outbox to `/api/sync`
Same MongoDB, same Render URL, same project	Same MongoDB, same Render URL, same project — nothing to re-create

Do this in the order given

Backend first, then the app. The new screens call endpoints that only exist in the updated backend, so an app-first update shows an error on every screen you just enabled.

Step	What you do	Time	Pass condition
0	Check which backend you are running	1 min	/api/schemes returns data, not 501
1	Replace the backend folder and redeploy on Render	10 min	health shows storeMode mongo
2	Copy 20 files into the Android project	3 min	Gradle syncs with no new errors
3	Clean build, uninstall the old APK, install the new one	5 min	Home shows 8 tiles
4	Walk the verification list	10 min	Every screen returns real data
5	Optional: enable AWS speech for the voice screen	10 min	health lists aws-polly

Step 0 — Confirm what is deployed

One request tells you whether the backend needs replacing. Open this in a browser or run it in a terminal:

```
curl https://YOUR-SERVICE.onrender.com/api/schemes

# old skeleton backend:
# {"success":false,"error":{"message":"Schemes API is not implemented yet",
# "code":"NOT_IMPLEMENTED"}}
# updated backend:
# {"success":true,"data":[ ... 10 schemes ... ],"count":10}
```

501 means you need Step 1. If you already see schemes, skip to Step 2 — your backend is current.

Step 1 — Replace the backend and redeploy

Render deploys from Git, so the update is a folder swap plus one push.

1. In your local repository, replace the whole `backend/` folder with the one in this pack. Delete the old folder rather than merging into it — the old `src/controllers/placeholder.controller.js` and its routes are gone.
2. Leave `backend/.env` alone if you keep secrets locally; on Render the values live in Dashboard > Environment, not in the repo.
3. Commit and push to the branch Render watches. The build runs `npm ci --omit=dev`, then `npm start`. It takes about three minutes.
4. Load the demo content (the API also seeds itself on boot when the collections are empty).
5. Confirm the health payload and one real endpoint.

```
cd backend && npm install && npm run seed # uses MONGODB_URI from backend/.env

curl https://YOUR-SERVICE.onrender.com/api/health
curl https://YOUR-SERVICE.onrender.com/api/schemes | head -c 200
```

Environment variables on the Render service

Variable	Status	Value / effect
MONGODB_URI	already set	Keep it. If it is missing, health reports storeMode memory and data resets on restart
NODE_ENV	set to production	Hides internal error detail in responses
PORT	automatic	Render injects it; the API defaults to 5000
CORS_ORIGIN	optional	Only for browser clients. Native Android sends no Origin header
AWS_REGION / AWS_ACCESS_KEY_ID / AWS_SECRET_ACCESS_KEY / AWS_S3_BUCKET	optional	Needed only for the Virtual Call screen (Step 5). Everything else works without them
SPEECH_PROVIDER	optional	auto (default) uses AWS when credentials exist, otherwise text only
BEDROCK_MODEL_ID	optional	Rephrases retrieved answers; empty keeps the extractive answer

> **No database migration needed**
> The update only adds collections (savedscholarships, reminders, smslogs, voicesessions) and fills schemes, scholarships and knowledge. Mongoose creates them on first write, so nothing you already have in Atlas is touched or dropped. `npm run seed -- --force` is the only destructive command and you do not need it.

Step 2 — Copy 20 files into the Android project

Commit or copy the project first; that is your rollback point.

```
cd android/RuralAssistant
git add -A && git commit -m "before coming-soon update"
```

Then copy the pack's `android/app` folder over your project's `app` folder, replacing what is there. The layout matches, so files land in the right packages.

```
# macOS / Linux
cp -R patch/android/app/. /path/to/RuralAssistant/app/

# Windows: select android\app, Ctrl+C, paste into ...\RuralAssistant\,
# choose "Replace the files in the destination"
```

New files (5)

File	What it adds
data/remote/api/FeatureApi.kt	Retrofit interface for every feature endpoint, plus the OkHttp client
ui/screens/features/FeatureScreens.kt	The working Assistant, Schemes, Scholarships, Reminders, Scam and SMS screens
ui/screens/call/VirtualCallScreen.kt	Records a clip, sends it as a voice turn, plays the reply
ui/screens/sms/SmsScreen.kt	Wraps the SMS demo screen with the saved language
utils/DeviceId.kt	One UUID per install, sent as X-Device-Id

Replaced files (15)

File	Why it changes
AndroidManifest.xml	Adds the RECORD_AUDIO permission for voice input
data/remote/api/RuralApiService.kt	Adds POST /api/sync for the outbox
ui/navigation/AppNavHost.kt	Routes the six placeholder screens to the real ones, plus SMS and Virtual Call
ui/navigation/Screen.kt	Adds the Sms and VirtualCall routes
ui/screens/home/HomeScreen.kt	Eight tiles instead of six
ui/screens/assistant, health	Thin wrappers that pass the saved language into the working screen
ui/screens/schemes, scholarships, reminders, scam	Thin wrappers, same pattern
worker/SyncWorker.kt	Actually posts the Room outbox and clears what the server accepts
res/values/strings.xml, values-hi, values-kn	Adds the SMS Centre and Virtual Call titles in all three languages

> **Do not copy Gradle files**
> The pack deliberately contains no `build.gradle.kts`, `libs.versions.toml` or wrapper. Your project already has Retrofit 2.11 with the Gson converter, OkHttp 4.12, Compose BOM 2024.10.01 and material-icons-extended — everything the new screens import. Replacing the Gradle files would only downgrade your toolchain.

Step 3 — Build, replace the old APK, run

A plain Run is not enough: Android Studio will happily reuse the previous install and you will keep seeing "Coming soon".

1. Set the backend URL in `android/RuralAssistant/local.properties` — HTTPS, with the trailing slash:

```
API_BASE_URL=https://YOUR-SERVICE.onrender.com/
```

2. **File > Sync Project with Gradle Files**, then **Build > Clean Project** and **Build > Rebuild Project**.
3. Uninstall the old app so the launcher and home screen are rebuilt from scratch:

```
adb uninstall com.ruralassistant.app
```

4. Run the app on the emulator or phone. Choose a language, then open **Settings > Check backend**: Connected, with the database state shown.
5. If the home screen still shows six tiles, the old APK is still installed — repeat steps 2 and 3.

Step 4 — Walk this list before you demo

Ten minutes, in this order. Each line names the thing that proves it worked.

1. **Settings > Check backend** — Connected, database *connected*, not *in-memory*.
2. **Ask** — a Kannada question, for example *a question about fever, asked in Kannada* — answer appears with a trusted source, and Read answer aloud speaks it.
3. **Health** — an English health question returns an answer, not a placeholder.
4. **Schemes** — search *farmer*, open PM-KISAN, tap Check general eligibility: a summary and a document checklist appear instead of raw JSON.
5. **Scholarships** — save one, confirm it moves to the saved state.
6. **Reminders** — add one with `2026-11-01T09:00:00+05:30`, mark it complete, delete it.
7. **Scam Checker** — paste a message asking for an OTP or KYC update: HIGH RISK with reasons and advice.
8. **SMS Centre** — valid number, tick the consent box: "Simulated — recorded only, not delivered".
9. **Settings > Sync now** — no crash, and the pending count clears when the network is up.
10. **Virtual Call** — Start call connects; without AWS it returns the text answer and says so in the status line, which is the expected result until Step 5.

Step 5 — Optional: turn on spoken replies

Only the Virtual Call screen needs this. Skip it and that screen still returns text.

- Create an S3 bucket for staging clips, for example `gramsahayak-voice-<yourname>`, with a one-day expiry rule on `voice-turns/`.

- Create an IAM user allowed exactly: `polly:SynthesizeSpeech`, `transcribe:StartTranscriptionJob`, `transcribe:GetTranscriptionJob`, and `s3:PutObject` plus `s3:DeleteObject` on `arn:aws:s3:::your-bucket/voice-turns/*`.
- Add `AWS_REGION=ap-south-1`, the key pair, `AWS_S3_BUCKET` and `SPEECH_PROVIDER=aws` to the Render environment, then redeploy.
- Re-check `/api/health`: `speech` should list `aws-transcribe` and `aws-polly`.

Troubleshooting

Symptom	Cause	Fix
App still says "Coming soon"	old APK reused	Clean + Rebuild, adb uninstall, reinstall
"Unable to connect to backend"	wrong API_BASE_URL	https URL with trailing slash; Sync Gradle after editing local.properties
HTTP 501 or NOT_IMPLEMENTED	old backend live	Step 1, then confirm /api/schemes returns data
Schemes or scholarships empty	collections not seeded	npm run seed, or restart the service (auto-seeds when empty)
Health says storeMode memory	MONGODB_URI missing on Render	Add it in Environment, redeploy — otherwise data resets
First request takes 30 s	Render free tier cold start	Open /api/health before presenting
Answer says "no verified information"	no knowledge match	Expected: it refuses to invent. Ask a health, scheme or scam question
Scam result stays blank	backend older than this pack	Step 1
Voice returns text only	no AWS speech config	Step 5, or accept text for the demo
Gradle: unresolved reference	a file was not copied	Re-copy all 20 files, then Sync
Reminder rejected as invalid date	not ISO 8601	Use 2026-11-01T09:00:00+05:30

Rolling back

Both halves are reversible and independent. Backend: redeploy the previous commit in Render (Manual Deploy > previous commit), which restores the skeleton routes — but the new app screens will then show errors, so roll back the app too. App: `git reset --hard <commit from Step 2>` and rebuild. No database change is involved either way.

> **What to say if a judge asks what changed**
> Six placeholder screens became live ones, the backend went from one working route to ten, and the offline outbox now actually syncs. The seed content is still marked unverified and answers still refuse to invent — both are deliberate, and both are visible in the code.